

A Stitch in Time: Learning Procedural Workflow via Self-Supervised Plackett–Luce Ranking

Chengan Che¹, Chao Wang¹, Xinyue Chen¹, Sophia Tsoka², Luis C. Garcia-Peraza-Herrera¹

¹Visual Understanding Research Group, Department of Informatics, King’s College London, UK

²Department of Informatics, King’s College London, UK

{chengan.che, chao.wang, xinyue.1.chen, sophia.tsoka, luis.c.garcia_peraza_herrera}@kcl.ac.uk

Abstract

Procedural activities, ranging from routine cooking to complex surgical operations, are highly structured sequences of actions performed in a specific temporal order. Despite the success of current self-supervised learning (SSL) methods on static images and short clips, these models often overlook the underlying sequential structure of such activities. We expose this lack of procedural awareness with a motivating experiment: models pretrained on forward and time-reversed sequences produce highly similar features, confirming that their representations are blind to the underlying procedural order. To address this shortcoming, we propose **PL-Stitch**, a self-supervised framework that harnesses the inherent temporal order of video frames as a powerful supervisory signal. Our approach integrates two novel probabilistic objectives based on the Plackett-Luce (PL) model. The primary PL objective trains the model to sort sampled frames chronologically, compelling it to learn the global workflow progression. The secondary objective, a spatio-temporal jigsaw loss, complements the learning by capturing fine-grained, cross-frame object correspondences. Our approach consistently achieves superior performance across five surgical and cooking benchmarks. Specifically, PL-Stitch yields significant gains in surgical phase recognition (e.g., +11.4 pp in k-NN accuracy on Cholec80) and cooking action segmentation (e.g., +5.7 pp in linear probing accuracy on Breakfast), demonstrating its effectiveness for procedural video representation learning. Code and models are available at <https://github.com/visurg-ai/PL-Stitch>.

1. Introduction

Many human activities, from daily cooking to surgical operations, are defined by a procedural workflow, a sequence of multi-step actions in a specific temporal order. This raises a critical question: do current self-supervised learning (SSL) methods actually learn this procedural logic, or do they

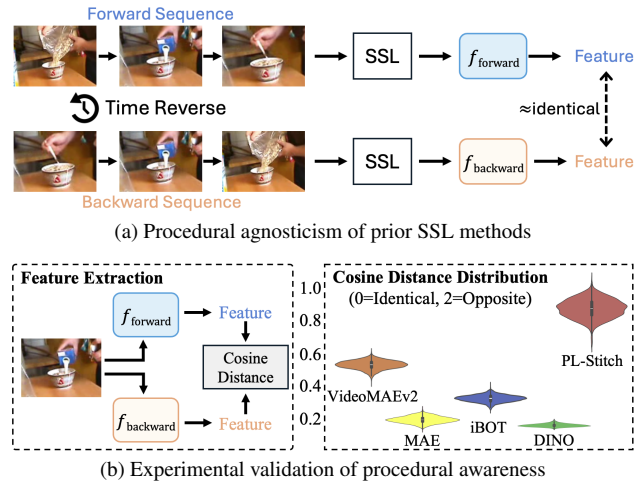


Figure 1. **Procedural awareness in self-supervised learning (SSL).** (a) Prior SSL methods are *procedural-agnostic*, learning features that fail to capture the procedural structure. (b) We pretrain PL-Stitch and prior SSL methods on the Breakfast dataset [23] using *forward* and *backward* sequences. The violin plot shows the cosine distance between the feature vectors of the same video frames. Baselines exhibit procedural agnosticism (low distance), whereas PL-Stitch’s high distance validates its procedural awareness.

merely recognize static actions?

To provide insights on this question, we conducted an experiment that motivates our work. We pretrained leading SSL models on both forward (chronological) and backward (time-reversed) video sequences of the Breakfast dataset [23]. At inference, we feed the same video frames to both the forward and backward trained encoders and compute the cosine distance between their resulting features. As shown in Fig. 1, these models produce nearly identical features regardless of temporal direction, confirming their representations are blind to the underlying procedural order. This demonstrates that while they can recognize an action (e.g., grinding beans), they fail to capture its essen-

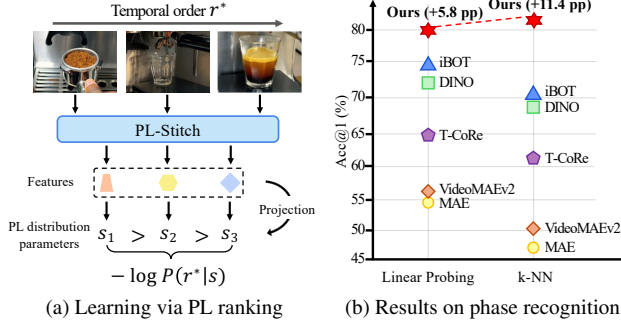


Figure 2. **Core concept of the PL-Stitch model and key results.** (a) Our model, PL-Stitch, learns a procedurally-aware representation by optimizing for the negative log likelihood of the Plackett-Luce distribution, $-\log P(r^*|s)$, which maximizes the probability (P) of the ground-truth temporal order (r^*) given the model’s predicted parameters (s). (b) Significant performance gains are shown for the Cholec80 phase recognition task.

tial temporal context (knowing it must come before brewing). This critical failure stems from the inherent design of prevalent SSL objectives, which are centered on local tasks like instance discrimination [4, 7, 8] or masked reconstruction [1, 18, 49]. Consequently, despite achieving strong performance in static image analysis [4, 7, 8, 18, 61] and short atomic clip modeling [13, 45, 49, 50], current SSL approaches remain fundamentally *procedural-agnostic* (Fig. 1a). They learn robust features for *what* is in a frame, but not *when* that frame occurs in the sequence.

To address this shortcoming, we propose **PL-Stitch**, a self-supervised framework that harnesses the inherent temporal order as a powerful supervisory signal (Fig. 2a). It is designed with two complementary branches.

The video branch serves as our main *temporal ranking objective*. It learns global workflow progression by tasking the model to predict the correct chronological order of sampled frames. Importantly, we formulate this as a listwise ranking problem using the probabilistic **Plackett-Luce (PL)** model [29, 34]. Our approach has **key advantages** over traditional temporal ordering tasks: 1) Its listwise formulation is more efficient and globally consistent than sub-optimal, pairwise comparisons [19, 24, 38], as it optimizes the ordering of a sequence of k elements in a single step, providing a global signal rather than relying on $\mathcal{O}(k^2)$ fragmented local comparisons. 2) Its probabilistic, ranking-based nature is a more robust fit for modeling relative order than classification tasks [15, 30, 54, 56]. Permutation classification tasks mistreat this relative problem by using absolute labels, penalizing near-correct orderings (minor sorting mistakes) as fully wrong. In contrast, our PL-based ranking model computes a probability distribution over permutations, allowing penalties to scale with error severity.

The complementary image branch learns fine-grained,

local features by jointly optimizing two objectives: 1) our novel spatio-temporal jigsaw objective, which uses adjacent past and future frames as temporal context to learn object correspondence, and 2) a masked image modeling (MIM) loss [61] for robust semantic representations.

Both video and image branch objectives are optimized jointly, compelling the encoder to learn a representation that is both semantically-rich and procedurally-aware. As shown in Fig. 2b, PL-Stitch consistently outperforms state-of-the-art methods using the same backbone.

In summary, our **contributions** are as follows:

- We identify and experimentally validate the procedural agnosticism of dominant SSL methods, demonstrating they are blind to the video’s underlying procedural order.
- To the best of our knowledge, we are the first to leverage the Plackett-Luce (PL) model to formulate probabilistic pretext tasks for self-supervised learning.
- We propose two novel objectives based on the PL model within our PL-Stitch framework: a listwise temporal ranking objective to learn global workflow progression and a spatio-temporal jigsaw objective to capture fine-grained object correspondence.
- We set a new state-of-the-art on five challenging surgical and cooking benchmarks, outperforming all baselines on both phase recognition and action segmentation tasks.

2. Related Work

Self-supervised Visual Representation Learning. A widely adopted approach in self-supervised visual representation learning is contrastive learning, which seeks to learn an embedding space that groups similar images together. Pioneer methods like MoCo [8, 9, 17] and SimCLR [7] learn this space by pulling augmented *positive* views of an image closer while pushing *negative* views apart. To prevent representational collapse without negative pairs, self-distillation methods like DINO [4] employ a student-teacher framework to match augmented views. Furthermore, these contrastive and distillation approaches are also adapted to the video domain by leveraging temporal dynamics [13, 35–37, 42, 48]. More recently, Masked Image Modeling (MIM) has become prominent, tasking a model with reconstructing masked patches at the pixel level (MAE [18]) or latent-space level (BEiT [1], iBOT [61]). Building on the iBOT objective, DINOv2 [31] and DINOv3 [39] later scaled this approach with massive data to boost performance. Masked Video Modeling (MVM) [14, 20, 33, 45, 49, 50, 55] extends this reconstruction paradigm to video. Methods like VideoMAE [45] learn representations by reconstructing randomly masked spatio-temporal tubes. However, a key limitation is its symmetric treatment of space and time [14, 20], which ignores the inherent causal progression of the temporal dimension and makes such methods suboptimal for learning the dynamics of procedural events.

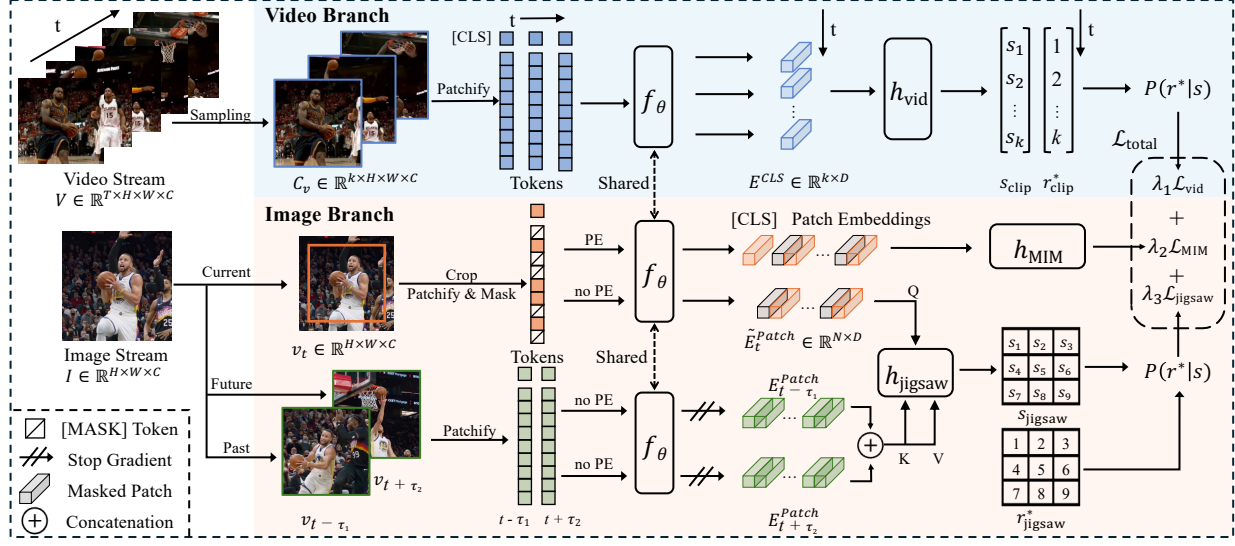


Figure 3. **Overview of the PL-Stitch framework.** Our model jointly trains a shared backbone encoder (f_θ) by using a **Video branch** (Sec. 3.4) for global workflow progression and an **Image branch** (Sec. 3.5) for fine-grained feature learning. The Video branch (top) treats time as order, training the encoder with a Plackett–Luce loss $\mathcal{L}_{vid}$ (Eq. 2, Eq. 3) to predict the correct relative chronological sequence of a sampled clip. The Image branch (bottom) learns robust local features by jointly optimizing a standard masked image modeling loss $\mathcal{L}_{MIM}$ with our novel spatio-temporal jigsaw $\mathcal{L}_{jigsaw}$, which learns object correspondence from adjacent frames (Eq. 4). The symbols h_{vid} , h_{MIM} , and h_{jigsaw} denote task-specific projection heads. By optimizing all objectives, the shared backbone learns a powerful representation sensitive to both procedural order and fine-grained visual details. **Best viewed online.**

Temporal Correspondence Learning. Understanding temporal correspondence between frames is crucial for modeling procedural activities. Early supervised methods using optical flow and motion estimation were effective but required costly pixel-level annotations [21, 41, 44, 58]. To overcome this issue, self-supervised approaches learned spatio-temporal correspondence by tracking objects or patches across frames [22, 26, 51, 57, 59]. More recently, temporal predictive reconstruction has emerged, where SiamMAE [16] predicts a future frame from the past and T-CoRe [28] using both past and future frames to reconstruct a central frame. While these methods excel at learning fine-grained correspondence, they lack a mechanism to capture video’s long-term procedural structure. Training models to order frame sequences has also been proposed [15, 19, 24, 30, 38, 54, 56]. However, this approach is limited by suboptimal objectives, such as pairwise comparisons that provide a local signal [19, 24, 38], or permutation classification tasks that miscast relative ordering as absolute classification [30, 54, 56]. Our work, PL-Stitch, addresses this gap by reformulating temporal understanding as a more robust, probabilistic listwise ranking problem, which we argue is a more natural and direct way to model sequences.

Procedural Activity Understanding. Modeling procedural activities, which involves both recognizing action steps and their temporal sequence, is a long-standing challenge in video understanding. Most recent works consider train-

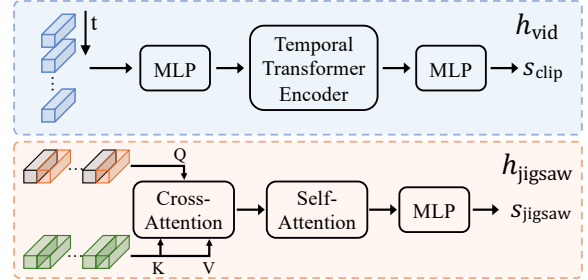


Figure 4. **Structure of our h_{vid} and h_{jigsaw} heads.** The video head (h_{vid}) consists of an MLP to reduce feature dimensionality for computational efficiency, a Transformer Encoder to aggregate global context across the k frame features for ordering, and a final MLP that outputs the PL distribution parameters s_{clip} . The jigsaw head (h_{jigsaw}) uses Cross-Attention to aggregate temporal context (K, V) onto the target patches (Q), followed by Self-Attention for spatial relationships refinement, and a final MLP for producing PL distribution parameters s_{jigsaw} .

ing on densely-annotated videos to localize and predict the order of steps [5, 10, 23, 43, 62]. To reduce this reliance on expensive labels, other methods have explored weakly-supervised approaches, such as learning from only a transcript of action steps without timestamps [3, 60, 62]. In contrast, our work operates in a fully self-supervised regime, learning a representation from the video signal alone that can be leveraged for procedural video understanding.

3. Methodology

3.1. Problem Formulation

Our primary objective is to pretrain a frame-level feature encoder, f_θ , without relying on manual annotations. An input video is a sequence $V = (v_1, v_2, \dots, v_T)$ of T frames, where each frame $v_t \in \mathbb{R}^{H \times W \times C}$. Following the standard Vision Transformer (ViT) paradigm, the encoder first tokenizes a frame v_t by splitting it into a grid of non-overlapping patches, each of size $P \times P$. These patches are first projected into D -dimensional patch tokens. A learnable [CLS] token is prepended to this sequence to form the initial input tokens, which we denote as $Z \in \mathbb{R}^{(N+1) \times D}$, where $N = H \times W / P^2$. The token sequence Z is then processed by the Transformer blocks to produce the final, context-aware output embeddings, denoted as $E \in \mathbb{R}^{(N+1) \times D}$. This completes the full mapping $f_\theta : \mathbb{R}^{H \times W \times C} \rightarrow \mathbb{R}^{(N+1) \times D}$.

3.2. Overview

Our goal is to learn a self-supervised representation capable of capturing both global procedural progression and fine-grained spatio-temporal relationships. To that end, we propose PL-Stitch, a multi-task framework that jointly trains a shared backbone encoder, f_θ , using two complementary branches as illustrated in Fig. 3. The **Video branch** (Sec. 3.4) is designed to learn temporal ordering across variable time scales by training the encoder to predict the correct chronological sequence of a sampled clip ($\mathcal{L}_{\text{vid}}$). In parallel, the **Image branch** (Sec. 3.5) learns robust local features through two tasks: our novel spatio-temporal jigsaw ($\mathcal{L}_{\text{jigsaw}}$), which learns fine-grained correspondence using adjacent frames as context and a masked image modeling objective ($\mathcal{L}_{\text{MIM}}$) based on iBOT [61]. Importantly, both the **temporal ordering** and the **spatio-temporal jigsaw** tasks are framed as listwise ranking problems, optimized with a unified **Plackett-Luce** objective (Sec. 3.3). The entire training procedure is summarized in Algorithm 1.

3.3. Plackett-Luce Ranking Formulation

Both our temporal and spatial learning tasks are framed as a listwise ranking problem. We address this using the Plackett-Luce (PL) distribution [29, 34], a probabilistic framework for defining a distribution over all $K!$ possible permutations of K items. The PL model is derived from Luce’s Choice Axiom [29] (independence from irrelevant alternatives), which states that the odds of choosing item i over item j depend only on those two items. This assumption aligns well with relative time ordering, where the preference between two frames is invariant to the rest of the sequence. Given a set of K items, the PL model is parameterized by a vector of real-valued scores $s \in \mathbb{R}^K$. The probability of observing a specific permutation r of the set

Algorithm 1 PL-Stitch Optimization Algorithm

Input: Unlabeled dataset $\mathcal{D}$.

Parameters: Batch size B , loss weights $\lambda_1, \lambda_2, \lambda_3$, and maximum iterations $L = \lfloor |\mathcal{D}| / B \rfloor$.

Initialize: Encoder f_θ and heads $h_{\text{vid}}, h_{\text{MIM}}, h_{\text{jigsaw}}$.

- 1: **procedure** PL-STITCH_STEP(V_m, I_m)
 - Video Branch*
 - 2: Sample sparse clip $C_v = (v_1, \dots, v_k)$ from V_m .
 - 3: Define target order $r_{\text{clip}}^* = (1, 2, \dots, k)$.
 - 4: Predict $s_{\text{clip}} = h_{\text{vid}}(f_\theta(C_v))$.
 - 5: Compute $\mathcal{L}_{\text{vid}} = -\log P(r_{\text{clip}}^* | s_{\text{clip}})$ by Eq. 3.
 - Image Branch*
 - 6: Sample frame triplet $(v_{t-\tau_1}, v_t, v_{t+\tau_2})$ from I_m and mask $v_t \rightarrow \tilde{v}_t$.
 - 7: Compute MIM loss $\mathcal{L}_{\text{MIM}}$ from $\tilde{v}_t, v_t$.
 - 8: Compute jigsaw query Q from $\tilde{v}_t$; context K, V from $v_{t-\tau_1}, v_{t+\tau_2}$ (no PE).
 - 9: Define target order $r_{\text{jigsaw}}^* = (1, 2, \dots, N)$.
 - 10: Predict $s_{\text{jigsaw}} = h_{\text{jigsaw}}(Q, K, V)$.
 - 11: Compute $\mathcal{L}_{\text{jigsaw}} = -\log P(r_{\text{jigsaw}}^* | s_{\text{jigsaw}})$ by Eq. 4.
 - 12: $\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{vid}} + \lambda_2 \mathcal{L}_{\text{MIM}} + \lambda_3 \mathcal{L}_{\text{jigsaw}}$ by Eq. 5.
 - 13: Update θ using $\nabla_\theta \mathcal{L}_{\text{total}}$.
 - 14: **end procedure**
 - 15: **for** $l = 1$ **to** L **do**
 - 16: Sample a batch of data $\{(V_i, I_i)\}_{i=1}^B$ from $\mathcal{D}$.
 - 17: **for** $m = 1$ **to** B **do**
 - 18: PL-STITCH_STEP(V_m, I_m) ▷ Parallelizable
 - 19: **end for**
 - 20: **end for**
 - 21: **return** Pre-trained encoder parameters θ .
-

$\{1, \dots, K\}$ is given by the likelihood:

$$P(r|s) = \prod_{i=1}^K \frac{\exp(s_{r(i)})}{\sum_{j=i}^K \exp(s_{r(j)})} \quad (1)$$

where $r(i)$ denotes the value of the item placed at position i in the permutation r . In our work, we train our self-supervised model to predict the vector of parameters s for a given set of K items (i.e., video frames for temporal ordering or image patches for jigsaw ranking). The network is trained via Maximum Likelihood Estimation (MLE) to learn parameters s that maximize the probability of the single correct ground-truth permutation r^* . This is achieved by minimizing the negative log-likelihood, which serves as our general ranking loss, $\mathcal{L}_{\text{PL}}$:

$$\mathcal{L}_{\text{PL}}(s, r^*) = -\log P(r^* | s). \quad (2)$$

This formulation is applied to both the video and jigsaw branches, providing a consistent optimization target for learning ordered sequences.

Table 3. **Ablation on components of PL-Stitch.** We report top-1 accuracy on linear probing and k -NN results for Cholec80 phase recognition.

No.	$\mathcal{L}_{\text{MIM}}$	$\mathcal{L}_{\text{vid}}$	$\mathcal{L}_{\text{jigsaw}}$	Linear	k -NN
1	✓			73.4	69.4
2	✓	✓		77.1	78.9
3	✓		✓	75.3	71.4
4	✓	✓	✓	77.8	80.2

Table 4. **Ablation on temporal objective formulation.** We replace our $\mathcal{L}_{\text{vid}}$ (PL) with simpler objectives (row 2 from Table 3).

No.	Temporal Objective	Linear	k -NN
1	Pairwise [19]	75.8	75.4
2	Permutation (CE) [30]	74.5	70.1
3	PL Ranking (Eq. 3)	77.1	78.9

Table 5. **Ablation on number of temporal frames (k) for $\mathcal{L}_{\text{vid}}$.**

No.	Frames (k)	Linear	k -NN
1	4	76.5	78.1
2	8	77.8	80.2
3	12	77.2	79.5
4	16	77.5	80.3

3.6. Total Objective

The final training objective for our PL-Stitch framework is the weighted sum of the three losses:

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{vid}} + \lambda_2 \mathcal{L}_{\text{MIM}} + \lambda_3 \mathcal{L}_{\text{jigsaw}} \quad (5)$$

where λ_1 , λ_2 , and λ_3 are hyperparameters balancing each task’s contribution. This final objective is used to update the network parameters during optimization (Algorithm 1).

4. Experiments

In this section, we first detail the experimental setup in Sec. 4.1. Next, we provide quantitative comparisons across five datasets in Sec. 4.2. Finally, we present ablation studies in Sec. 4.3 and qualitative results in Sec. 4.4 (see the supplementary material for more details).

4.1. Experimental Setup

Datasets and Evaluation Protocols. We evaluate on five widely used datasets from the surgical and cooking domains, selected for their long-range, multi-step procedural workflows with fixed sequential dependencies. For surgical procedures, we evaluate the temporal phase recognition task using AutoLaparo [52] (laparoscopic hysterectomy), along with Cholec80 [46] and M2CAI16 [40] (both laparoscopic cholecystectomy). For cooking activities, we evaluate the temporal action segmentation task using GTEA [12] (egocentric cooking) and Breakfast [23] (third-person cooking).

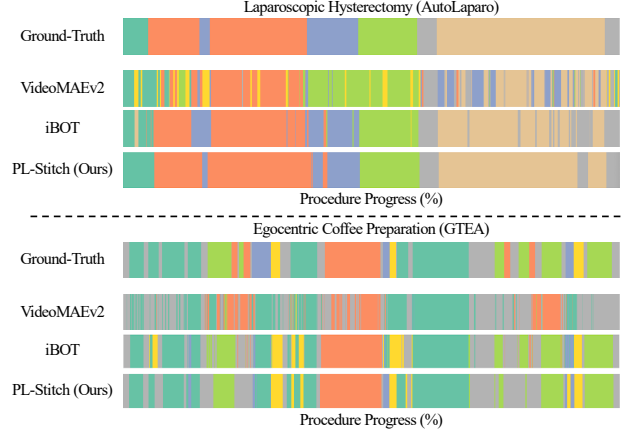


Figure 5. **Visualization of linear probing** predictions for phase recognition (top) and action segmentation (bottom). Each horizontal bar shows the frame-wise predictions, where the x-axis denotes progress over time and the colors represent different classes.

We assess feature quality using two standard protocols: **Linear Probing**, which trains a linear classifier on frozen features, and **k -NN Classification** [4], which uses $k = 20$ nearest neighbors in the feature space for non-parametric evaluation. For k -NN, we report top-1 accuracy. For Linear probing, we follow established benchmarks to report frame-wise accuracy and F1-score for surgical datasets [6, 27, 52], and frame-wise accuracy, segmental edit distance, and the segmental F1 score at overlapping thresholds 10%, 25% and 50% for cooking datasets [12, 23, 25].

Pretraining Details. We adopt ViT-B/16 [11] as our framework’s backbone. For our temporal ranking objective ($\mathcal{L}_{\text{vid}}$), we sample $k=8$ frames from each video of length T . We first select a random starting frame t_0 and then a random, uniform step length Δt , such that $t_0 + 7\Delta t \leq T$. Videos are resampled multiple times per epoch, providing broad coverage of the procedural structure. For the $\mathcal{L}_{\text{jigsaw}}$ objective, the past and future context frames are sampled from the temporal ranges $[-2.5, -1.5]$ s and $[+1.5, +2.5]$ s relative to the current frame, following [28]. For the $\mathcal{L}_{\text{MIM}}$ objective, we apply block-wise masking with a ratio of 30% as in [61]. The loss weights (Eq. 5) are set to $\lambda_1 = 1$, $\lambda_2 = 1$, and $\lambda_3 = 0.4$ with analysis in supplementary material.

Optimization Strategies. We employ different pretraining optimization strategies based on the target domain. For surgical datasets, models are pretrained for 30 epochs on the large-scale surgical video dataset LEMON [6] to leverage its domain-specific knowledge. We use AdamW with a base learning rate of 4×10^{-4} , a 3-epoch cosine warmup, and a total batch size of 240. For cooking datasets, we pretrain directly on their respective official training sets for 100 epochs with 10 warmup epochs, as no comparable large-scale pre-training dataset exists for this highly variable domain; other

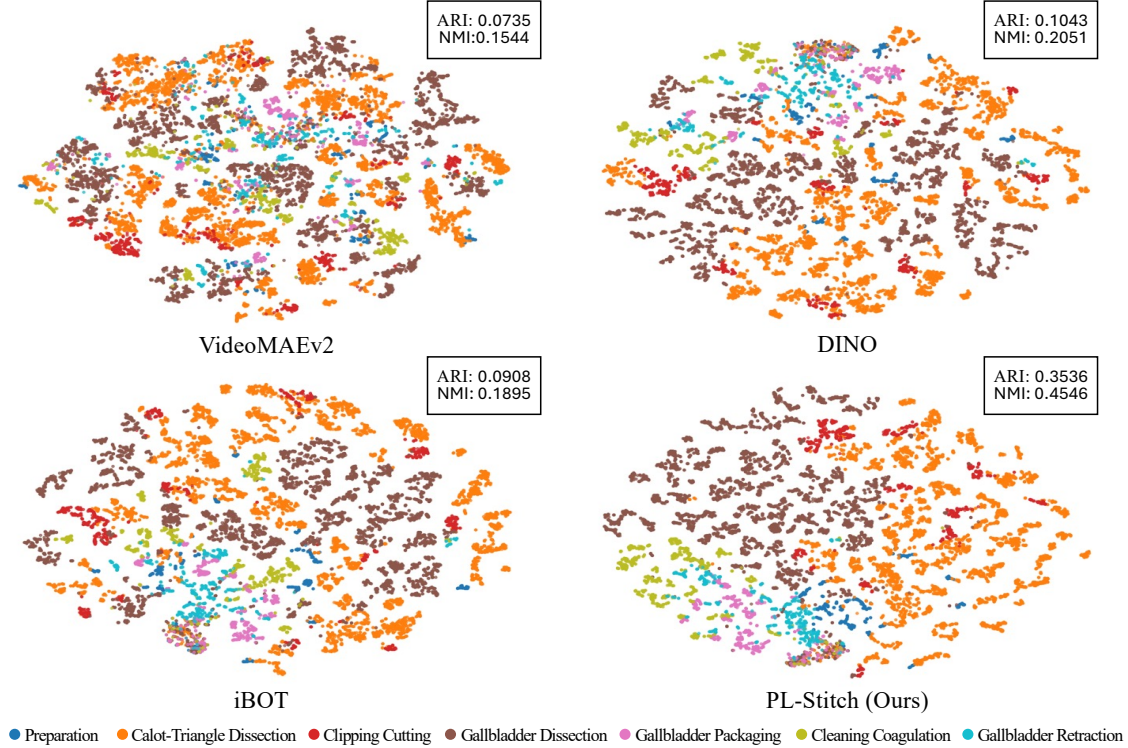


Figure 6. **t-SNE visualization** of frozen backbone features for Cholec80 phase recognition. The plot also reports the clustering quality metrics Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI), where higher values are better. Our PL-Stitch model demonstrates superior class separation (ARI: 0.3536, NMI: 0.4546) compared to VideoMAEv2, DINO, and iBOT.

optimizer settings are kept identical. Experiments utilize 4 NVIDIA A100 (40GB) GPUs and PyTorch 2.5.1 [32].

4.2. Main Quantitative Results

We provide a comprehensive comparison of PL-Stitch against leading self-supervised models.

Surgical Phase Recognition. As shown in Table 1, PL-Stitch consistently outperforms all baselines on AutoLaparo, Cholec80, and M2CAI16 under both linear probing and k -NN evaluation. This demonstrates superior feature quality over both generalist (G) and specialist (S) methods. The performance gains are particularly striking in the k -NN evaluation, which directly assesses feature space quality. On Cholec80, PL-Stitch achieves 81.7% k -NN accuracy, a significant gain of **11.4pp** over the strong iBOT baseline [61]. This trend holds for AutoLaparo (+7.2pp) and M2CAI16 (+9.1pp) as well. Furthermore, PL-Stitch also achieves the highest scores in all linear probing metrics, underscoring the discriminative power of its frozen features.

Cooking Action Segmentation. We validate our model’s generalizability on the cooking datasets in Table 2. On GTEA, PL-Stitch surpasses the best baselines in k -NN accuracy (62.4% vs. 60.0%) and in linear probing, with higher accuracy (54.1% vs. 52.2%), Edit score (60.2% vs. 57.2%),

and F1@10 (61.7% vs. 59.3%). On Breakfast, this superiority is even more pronounced with a linear probing accuracy gain of **+5.7pp** over the second-best method (DINO).

4.3. Ablation Study

We conduct ablations on the Cholec80 phase recognition task by pretraining on LEMON [6] and report top-1 accuracy under linear probing and k -NN ($k = 20$). We use a ViT-S/16 backbone to reduce computational cost. Our default settings in Tables 3, 4, and 5 are highlighted in gray.

Effect of PL-Stitch Components. We analyze the contribution of each objective in Table 3. Our baseline (row 1), with only the $\mathcal{L}_{\text{MIM}}$ objective [61], achieves 69.4% k -NN accuracy. Adding our core temporal ranking objective ($\mathcal{L}_{\text{vid}}$) (row 2) provides the largest single boost, improving k -NN accuracy by **9.5pp** to 78.9%. Adding only the spatio-temporal jigsaw objective ($\mathcal{L}_{\text{jigsaw}}$) (row 3) provides a smaller benefit. Finally, our full PL-Stitch model (row 4), combining all three objectives, achieves the highest accuracy at **80.2%**. This demonstrates that our global temporal and local jigsaw objectives are complementary, and their joint optimization leads to the most robust representation.

Impact of PL Ranking Formulation. We compare our Plackett-Luce (PL) formulation against traditional ordering

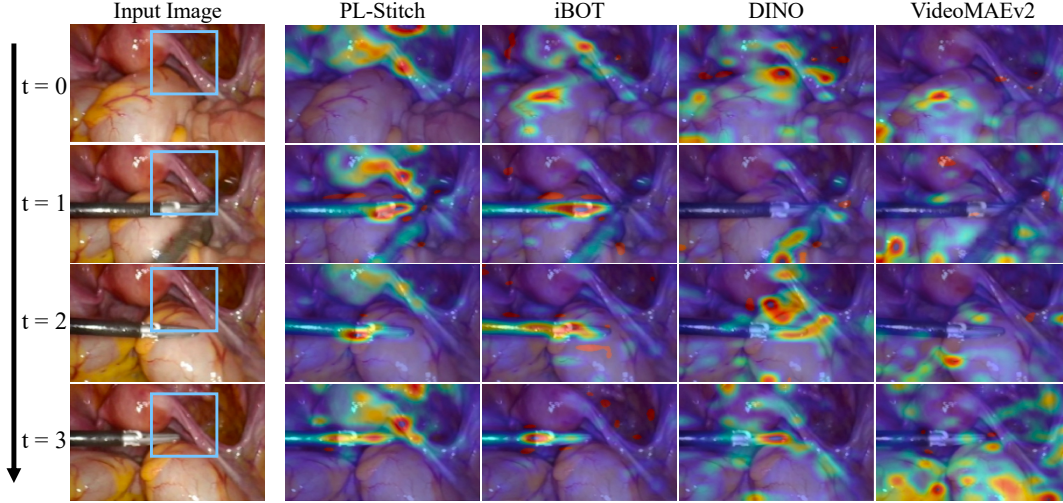


Figure 7. **Attention maps** queried by the [CLS] token comparing our method, PL-Stitch, against DINO, iBOT, and VideoMAEv2 on a sequence of frames from AutoLaparo [52] during the *Dividing Ligament and Peritoneum* phase. The light blue box indicates the instrument operating area, which is critical for understanding the surgical phase. While other methods produce diffuse attention that changes erratically over time, our PL-Stitch maintains a consistent and accurate focus on the target area (the instrument and its operating area).

objectives (Table 4). We replace our $\mathcal{L}_{\text{vid}}$ with two common baselines: 1) a Pairwise Loss [19], which enforces the order of all pairs in the sequence, and 2) a Permutation Classification loss [30], which treats each shuffled permutation of the sequence as a distinct class. Both baselines perform worse than our PL objective, confirming that the probabilistic, listwise formulation of the PL model is key to learning a robust, progress-aware representation.

Number of Temporal Ranking Frames (k). We analyze the number of frames k used in the temporal ranking objective ($\mathcal{L}_{\text{vid}}$) in Table 5. Performance on the k -NN evaluation improves by 2.1 pp when increasing k from 4 to 8, as this provides a richer temporal context for ranking. While performance at $k = 12$ and $k = 16$ is similar to $k = 8$ (77.5% vs 77.8% on linear probing, 80.3% vs 80.2% on k -NN), increasing the sequence length raises the computational burden (up to $4\times$ for $k = 16$). We therefore selected $k = 8$ as the optimal balance between accuracy and efficiency.

4.4. Qualitative Results

Temporal Consistency. As shown in Fig. 5, linear probe predictions from PL-Stitch on both AutoLaparo and GTEA are significantly more stable and temporally consistent than the competing methods, demonstrating PL-Stitch’s ability to learn a robust procedural representation.

Feature Space Visualization. Fig. 6 visualizes the frozen Cholec80 features using t-SNE [47]. Baseline features from VideoMAEv2, DINO, and iBOT overlap heavily, showing no clear phase separation. In contrast, PL-Stitch forms distinct, well-separated clusters that align with ground-truth phases. Its clustering quality (ARI : 0.3536, NMI : 0.4546)

is more than twice as high as the second-best baseline, DINO (ARI : 0.1043, NMI : 0.2051). This clear separation visually explains its high k -NN accuracy and confirms its learned procedural awareness.

Attention Maps. We choose the [CLS] token as the query and visualize its resulting attention map in Fig. 7. The baselines show scattered and unstable attention, failing to track the surgical action. In contrast, our PL-Stitch maintains a stable and precise focus on the instrument and its operating area. This persistent localization provides qualitative evidence that our model learns the procedural context, not just static objects. Additional qualitative examples are in the supplementary material.

5. Conclusion

We addressed the procedural agnosticism of modern self-supervised learning methods, first demonstrating with a motivating experiment that they are blind to procedural order. To solve this, we proposed PL-Stitch, a novel self-supervised framework that harnesses the video’s inherent temporal order as a powerful supervisory signal. By integrating probabilistic Plackett-Luce ranking for both global workflow progression and fine-grained object correspondence, PL-Stitch sets a new state-of-the-art on challenging surgical and cooking benchmarks. This validates our core insight that explicitly modeling temporal order is key to learning procedurally-aware video representations. Future work will move beyond representation learning to generative tasks, such as action anticipation, and explore multi-modal integration by aligning the learned procedural steps with instructional text from recipes or surgical manuals.

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References

- [1] Hangbo Bao, Li Dong, Songhao Piao, and Furu Wei. BEiT: BERT Pre-Training of Image Transformers. In *International Conference on Learning Representations*, 2022. 2
- [2] Dominik Batić, Felix Holm, Ege Özsoy, Tobias Czempel, and Nassir Navab. EndoViT: pretraining vision transformers on a large collection of endoscopic images. *International Journal of Computer Assisted Radiology and Surgery*, 19(6): 1085–1091, 2024. 5
- [3] P. Bojanowski, R. Lajugie, E. Grave, F. Bach, I. Laptev, J. Ponce, and C. Schmid. Weakly-Supervised Alignment of Video with Text. In *2015 IEEE International Conference on Computer Vision (ICCV)*, pages 4462–4470. IEEE, 2015. 3
- [4] Mathilde Caron, Hugo Touvron, Ishan Misra, Herve Jegou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging Properties in Self-Supervised Vision Transformers. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 9630–9640. IEEE, 2021. 2, 5, 6
- [5] Chien-Yi Chang, De-An Huang, Danfei Xu, Ehsan Adeli, Li Fei-Fei, and Juan Carlos Niebles. Procedure Planning in Instructional Videos. *Computer Vision – ECCV 2020*, pages 334–350, 2020. 3
- [6] Chengan Che, Chao Wang, Tom Vercauteren, Sophia Tsoka, and Luis C. Garcia-Peraza-Herrera. LEMON: A Large Endoscopic MONocular Dataset and Foundation Model for Perception in Surgical Settings. *arXiv preprint arXiv:2503.19740*, 2025. 5, 6, 7
- [7] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In *Proceedings of the 37th International Conference on Machine Learning*. JMLR.org, 2020. 2
- [8] Xinlei Chen, Haoqi Fan, Ross Girshick, and Kaiming He. Improved Baselines with Momentum Contrastive Learning. *arXiv preprint arXiv:2003.04297*, 2020. 2
- [9] Xinlei Chen, Saining Xie, and Kaiming He. An Empirical Study of Training Self-Supervised Vision Transformers. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 9620–9629. IEEE, 2021. 2
- [10] Dima Damen, Hazel Doughty, Giovanni Maria Farinella, Sanja Fidler, Antonino Furnari, Evangelos Kazakos, Davide Moltisanti, Jonathan Munro, Toby Perrett, Will Price, and Michael Wray. The EPIC-KITCHENS Dataset: Collection, Challenges and Baselines. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43(11):4125–4141, 2021. 3
- [11] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. *arXiv preprint arXiv:2010.11929*, abs/2010.11929, 2020. 6
- [12] Alireza Fathi, Xiaofeng Ren, and James M. Rehg. Learning to recognize objects in egocentric activities. In *CVPR 2011*, pages 3281–3288. IEEE, 2011. 5, 6
- [13] Christoph Feichtenhofer, Haoqi Fan, Bo Xiong, Ross Girshick, and Kaiming He. A Large-Scale Study on Unsupervised Spatiotemporal Representation Learning. In *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3298–3308. IEEE, 2021. 2
- [14] Christoph Feichtenhofer, haoqi fan, Yanghao Li, and Kaiming He. Masked Autoencoders As Spatiotemporal Learners. In *Advances in Neural Information Processing Systems*, pages 35946–35958. Curran Associates, Inc., 2022. 2
- [15] Basura Fernando, Hakan Bilen, Efstratios Gavves, and Stephen Gould. Self-Supervised Video Representation Learning with Odd-One-Out Networks. In *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5729–5738. IEEE, 2017. 2, 3
- [16] Agrim Gupta, Jiajun Wu, Jia Deng, and Li Fei-Fei. Siamese masked autoencoders. In *Proceedings of the 37th International Conference on Neural Information Processing Systems*, Red Hook, NY, USA, 2023. Curran Associates Inc. 3
- [17] Kaiming He, Haoqi Fan, Yuxin Wu, Saining Xie, and Ross Girshick. Momentum Contrast for Unsupervised Visual Representation Learning. In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 9726–9735. IEEE, 2020. 2
- [18] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollar, and Ross Girshick. Masked Autoencoders Are Scalable Vision Learners. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 15979–15988. IEEE, 2022. 2, 5
- [19] Kai Hu, Jie Shao, Yuan Liu, Bhiksha Raj, Marios Savvides, and Zhiqiang Shen. Contrast and Order Representations for Video Self-supervised Learning. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 7919–7929. IEEE, 2021. 2, 3, 6, 8
- [20] Bingkun Huang, Zhiyu Zhao, Guozhen Zhang, Yu Qiao, and Limin Wang. MGMAE: Motion Guided Masking for Video Masked Autoencoding. In *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 13447–13458. IEEE, 2023. 2
- [21] Eddy Ilg, Nikolaus Mayer, Tonmoy Saikia, Margret Keuper, Alexey Dosovitskiy, and Thomas Brox. FlowNet 2.0: Evolution of Optical Flow Estimation with Deep Networks. In *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1647–1655. IEEE, 2017. 3
- [22] Allan A Jabri, Andrew Owens, and Alexei A Efros. Space-time correspondence as a contrastive random walk. In *Proceedings of the 34th International Conference on Neural Information Processing Systems*, Red Hook, NY, USA, 2020. Curran Associates Inc. 3
- [23] Hilde Kuehne, Ali Arslan, and Thomas Serre. The Language of Actions: Recovering the Syntax and Semantics of Goal-Directed Human Activities. In *2014 IEEE Conference on Computer Vision and Pattern Recognition*, pages 780–787. IEEE, 2014. 1, 3, 5, 6

- [24] Hsin-Ying Lee, Jia-Bin Huang, Maneesh Singh, and Ming-Hsuan Yang. Unsupervised Representation Learning by Sorting Sequences. In *2017 IEEE International Conference on Computer Vision (ICCV)*, pages 667–676. IEEE, 2017. 2, 3
- [25] Shijie Li, Yazan Abu Farha, Yun Liu, Ming-Ming Cheng, and Juergen Gall. MS-TCN++: Multi-Stage Temporal Convolutional Network for Action Segmentation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(6):6647–6658, 2023. 6
- [26] Xueting Li, Sifei Liu, Shalini De Mello, Xiaolong Wang, Jan Kautz, and Ming-Hsuan Yang. Joint-task self-supervised learning for temporal correspondence. In *Proceedings of the 33rd International Conference on Neural Information Processing Systems*. Curran Associates Inc., Red Hook, NY, USA, 2019. 3
- [27] Yang Liu, Maxence Boels, Luis C. Garcia-Peraza-Herrera, Tom Vercauteren, Prokar Dasgupta, Alejandro Granados, and Sébastien Ourselin. LoViT: Long Video Transformer for surgical phase recognition. *Medical Image Analysis*, 99: 103366, 2025. 6
- [28] Yang Liu, Qianqian Xu, Peisong Wen, Siran Dai, and Qingming Huang. When the Future Becomes the Past: Taming Temporal Correspondence for Self-supervised Video Representation Learning. In *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 24033–24044. IEEE, 2025. 3, 5, 6
- [29] R.Duncan Luce. The choice axiom after twenty years. *Journal of Mathematical Psychology*, 15(3):215–233, 1977. 2, 4
- [30] Ishan Misra, C Lawrence Zitnick, and Martial Hebert. Shuffle and Learn: Unsupervised Learning Using Temporal Order Verification. In *Computer Vision – ECCV 2016*, pages 527–544, Cham, 2016. Springer International Publishing. 2, 3, 6, 8
- [31] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Hervé Jegou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. DINOv2: Learning Robust Visual Features without Supervision. *arXiv preprint arXiv:2304.07193*, 2023. 2
- [32] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, Alban Desmaison, Andreas Köpf, Edward Yang, Zach DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, Junjie Bai, and Soumith Chintala. PyTorch: an imperative style, high-performance deep learning library. In *Proceedings of the 33rd International Conference on Neural Information Processing Systems*. Curran Associates Inc., Red Hook, NY, USA, 2019. 7
- [33] Gensheng Pei, Tao Chen, Xiruo Jiang, Huafeng Liu, Zeren Sun, and Yazhou Yao. VideoMAC: Video Masked Autoencoders Meet ConvNets. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 22733–22743. IEEE, 2024. 2
- [34] R L Plackett. The Analysis of Permutations. *Journal of the Royal Statistical Society. Series C (Applied Statistics)*, 24(2): 193–202, 1975. 2, 4
- [35] Rui Qian, Tianjian Meng, Boqing Gong, Ming-Hsuan Yang, Huisheng Wang, Serge Belongie, and Yin Cui. Spatiotemporal Contrastive Video Representation Learning. In *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6960–6970. IEEE, 2021. 2
- [36] Zhiwu Qing, Shiwei Zhang, Ziyuan Huang, Yi Xu, Xiang Wang, Changxin Gao, Rong Jin, and Nong Sang. Self-Supervised Learning from Untrimmed Videos via Hierarchical Consistency. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(10):12408–12426, 2023.
- [37] Adria Recasens, Pauline Luc, Jean-Baptiste Alayrac, Luyu Wang, Florian Strub, Corentin Tallec, Mateusz Malinowski, Viorica Patraucean, Florent Altche, Michal Valko, Jean-Bastien Grill, Aaron van den Oord, and Andrew Zisserman. Broaden Your Views for Self-Supervised Video Learning. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 1235–1245. IEEE, 2021. 2
- [38] Pierre Sermanet, Corey Lynch, Jasmine Hsu, and Sergey Levine. Time-Contrastive Networks: Self-Supervised Learning from Multi-view Observation. In *2017 IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 486–487. IEEE, 2017. 2, 3
- [39] Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, Marc Szafraniec, Seungeun Yi, Michaël Ramamonjisoa, Francisco Massa, Daniel Haziza, Luca Wehrstedt, Jianyuan Wang, Timothée Darcet, Théo Moutakanni, Leonel Sentana, Claire Roberts, Andrea Vedaldi, Jamie Tolan, John Brandt, Camille Couprie, Julien Mairal, Hervé Jégou, Patrick Labatut, and Piotr Bojanowski. DINOv3. *arXiv preprint arXiv:2508.10104*, 2025. 2
- [40] Ralf Stauder, Daniel Ostler, Michael Kranzfelder, Sebastian Koller, Hubertus Feußner, and Nassir Navab. The TUM LapChole dataset for the M2CAI 2016 workflow challenge. *arXiv preprint arXiv:1610.09278*, 2016. 6
- [41] Deqing Sun, Xiaodong Yang, Ming-Yu Liu, and Jan Kautz. PWC-Net: CNNs for Optical Flow Using Pyramid, Warping, and Cost Volume. In *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 8934–8943. IEEE, 2018. 3
- [42] Xinyu Sun, Peihao Chen, Liangwei Chen, Changhao Li, Thomas H. Li, Minghui Tan, and Chuang Gan. Masked Motion Encoding for Self-Supervised Video Representation Learning. In *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2235–2245. IEEE, 2023. 2
- [43] Yansong Tang, Dajun Ding, Yongming Rao, Yu Zheng, Danyang Zhang, Lili Zhao, Jiwen Lu, and Jie Zhou. COIN: A Large-Scale Dataset for Comprehensive Instructional Video Analysis. In *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1207–1216. IEEE, 2019. 3

- [44] Zachary Teed and Jia Deng. RAFT: Recurrent All-Pairs Field Transforms for Optical Flow. In *Computer Vision – ECCV 2020*, pages 402–419, Cham, 2020. Springer International Publishing. 3
- [45] Zhan Tong, Yibing Song, Jue Wang, and Limin Wang. VideoMAE: masked autoencoders are data-efficient learners for self-supervised video pre-training. In *Proceedings of the 36th International Conference on Neural Information Processing Systems*, Red Hook, NY, USA, 2022. Curran Associates Inc. 2
- [46] Andru P. Twinanda, Sherif Shehata, Didier Mutter, Jacques Marescaux, Michel de Mathelin, and Nicolas Padoy. EndoNet: A Deep Architecture for Recognition Tasks on Laparoscopic Videos. *IEEE Transactions on Medical Imaging*, 36(1):86–97, 2017. 6
- [47] Laurens Van Der Maaten. Accelerating t-SNE using tree-based algorithms. *J. Mach. Learn. Res.*, 15(1):3221–3245, 2014. 8
- [48] Jinpeng Wang, Yuting Gao, Ke Li, Jianguo Hu, Xinyang Jiang, Xiaowei Guo, Rongrong Ji, and Xing Sun. Enhancing Unsupervised Video Representation Learning by Decoupling the Scene and the Motion. *Proceedings of the AAAI Conference on Artificial Intelligence*, 35(11):10129–10137, 2021. 2
- [49] Limin Wang, Bingkun Huang, Zhiyu Zhao, Zhan Tong, Yinan He, Yi Wang, Yali Wang, and Yu Qiao. VideoMAE V2: Scaling Video Masked Autoencoders with Dual Masking. In *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 14549–14560. IEEE, 2023. 2, 5
- [50] Rui Wang, Dongdong Chen, Zuxuan Wu, Yinpeng Chen, Xiyang Dai, Mengchen Liu, Yu-Gang Jiang, Luowei Zhou, and Lu Yuan. BEVT: BERT Pretraining of Video Transformers. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 14713–14723. IEEE, 2022. 2
- [51] Xiaolong Wang, Allan Jabri, and Alexei A. Efros. Learning Correspondence From the Cycle-Consistency of Time. In *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2561–2571. IEEE, 2019. 3
- [52] Ziyi Wang, Bo Lu, Yonghao Long, Fangxun Zhong, Tak-Hong Cheung, Qi Dou, and Yunhui Liu. AutoLaparo: A New Dataset of Integrated Multi-tasks for Image-guided Surgical Automation in Laparoscopic Hysterectomy. In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2022*, pages 486–496, Cham, 2022. Springer Nature Switzerland. 6, 8
- [53] Zhao Wang, Chang Liu, Shaoting Zhang, and Qi Dou. Foundation Model for Endoscopy Video Analysis via Large-Scale Self-supervised Pre-train. In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2023*, pages 101–111, Cham, 2023. Springer Nature Switzerland. 5
- [54] Donglai Wei, Joseph Lim, Andrew Zisserman, and William T Freeman. Learning and Using the Arrow of Time. In *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 8052–8060. IEEE, 2018. 2, 3
- [55] Qiangqiang Wu, Tianyu Yang, Ziquan Liu, Baoyuan Wu, Ying Shan, and Antoni B. Chan. DropMAE: Masked Autoencoders with Spatial-Attention Dropout for Tracking Tasks. In *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 14561–14571. IEEE, 2023. 2
- [56] Dejing Xu, Jun Xiao, Zhou Zhao, Jian Shao, Di Xie, and Yueting Zhuang. Self-Supervised Spatiotemporal Learning via Video Clip Order Prediction. In *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10326–10335. IEEE, 2019. 2, 3
- [57] Jiarui Xu and Xiaolong Wang. Rethinking Self-supervised Correspondence Learning: A Video Frame-level Similarity Perspective. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 10055–10065. IEEE, 2021. 3
- [58] Jia Xu, Rene Ranftl, and Vladlen Koltun. Accurate Optical Flow via Direct Cost Volume Processing. In *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5807–5815. IEEE, 2017. 3
- [59] Yurong Zhang, Liulei Li, Wenguan Wang, Rong Xie, Li Song, and Wenjun Zhang. Boosting Video Object Segmentation via Space-Time Correspondence Learning. In *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2246–2256. IEEE, 2023. 3
- [60] He Zhao, Isma Hadji, Nikita Dvornik, Konstantinos G. Derpanis, Richard P. Wildes, and Allan D. Jepson. P³IV: Probabilistic Procedure Planning from Instructional Videos with Weak Supervision. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2928–2938. IEEE, 2022. 3
- [61] Jinghao Zhou, Chen Wei, Huiyu Wang, Wei Shen, Cihang Xie, Alan Yuille, and Tao Kong. iBOT: Image BERT Pre-Training with Online Tokenizer. *International Conference on Learning Representations (ICLR)*, 2022. 2, 4, 5, 6, 7
- [62] Dimitri Zhukov, Jean-Baptiste Alayrac, Ramazan Gokberk Cinbis, David Fouhey, Ivan Laptev, and Josef Sivic. Cross-Task Weakly Supervised Learning From Instructional Videos. In *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3532–3540. IEEE, 2019. 3